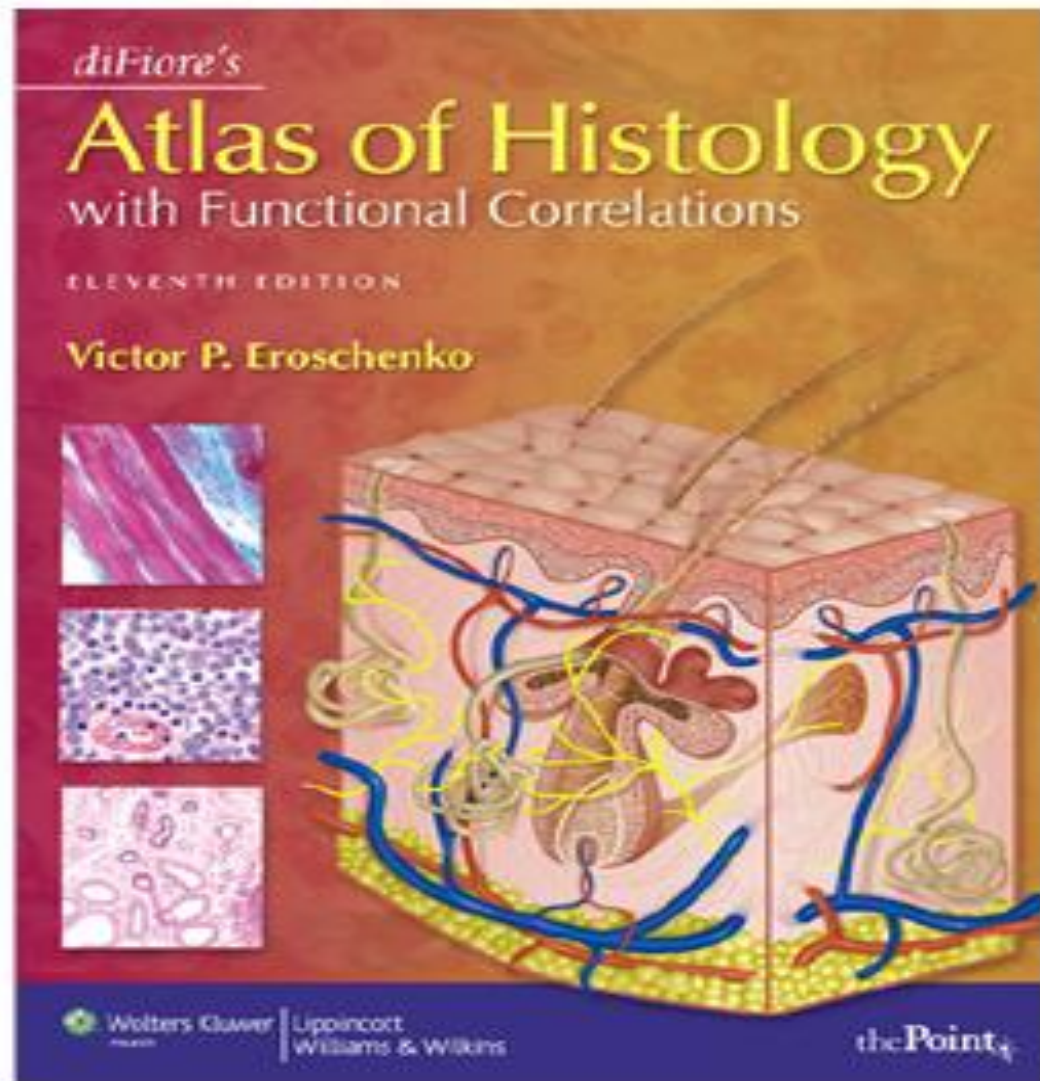
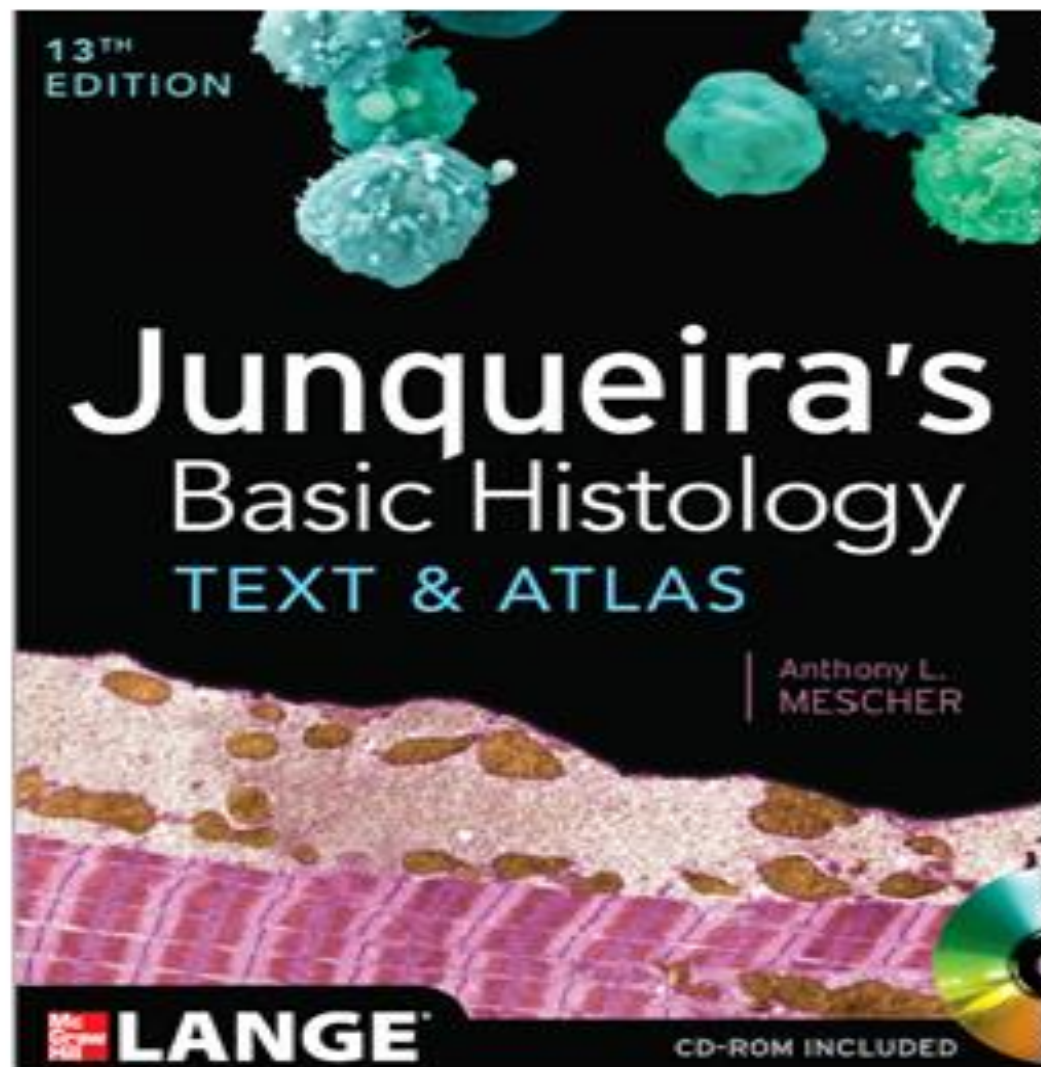


Histology





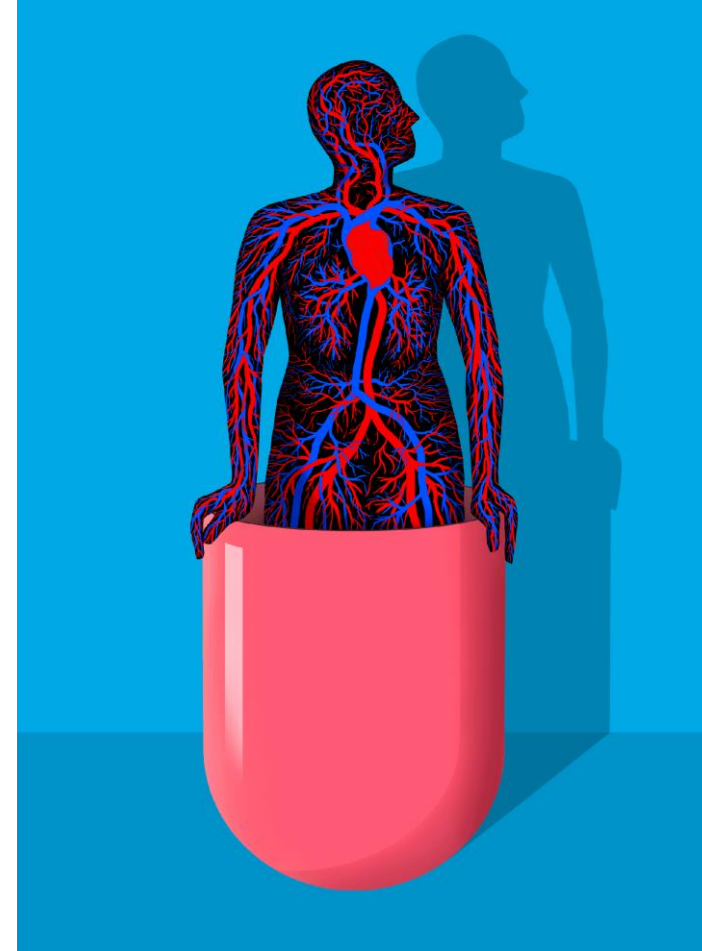
Alameed university-Faculty of medicine

Histology of the Circulatory System L1

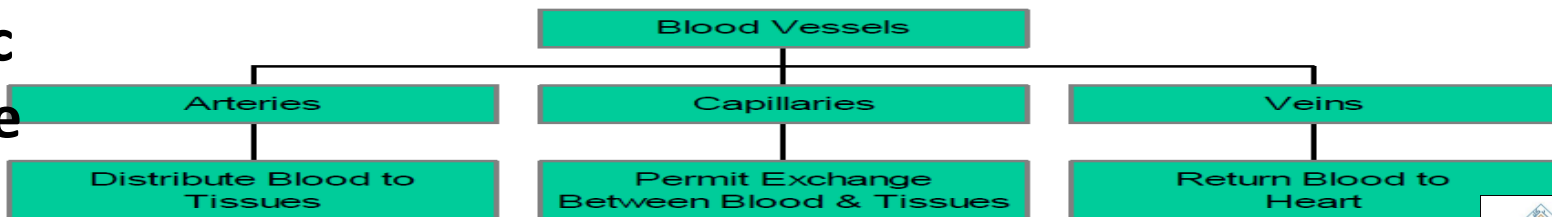
Dr. Shehab Ahmad Faris

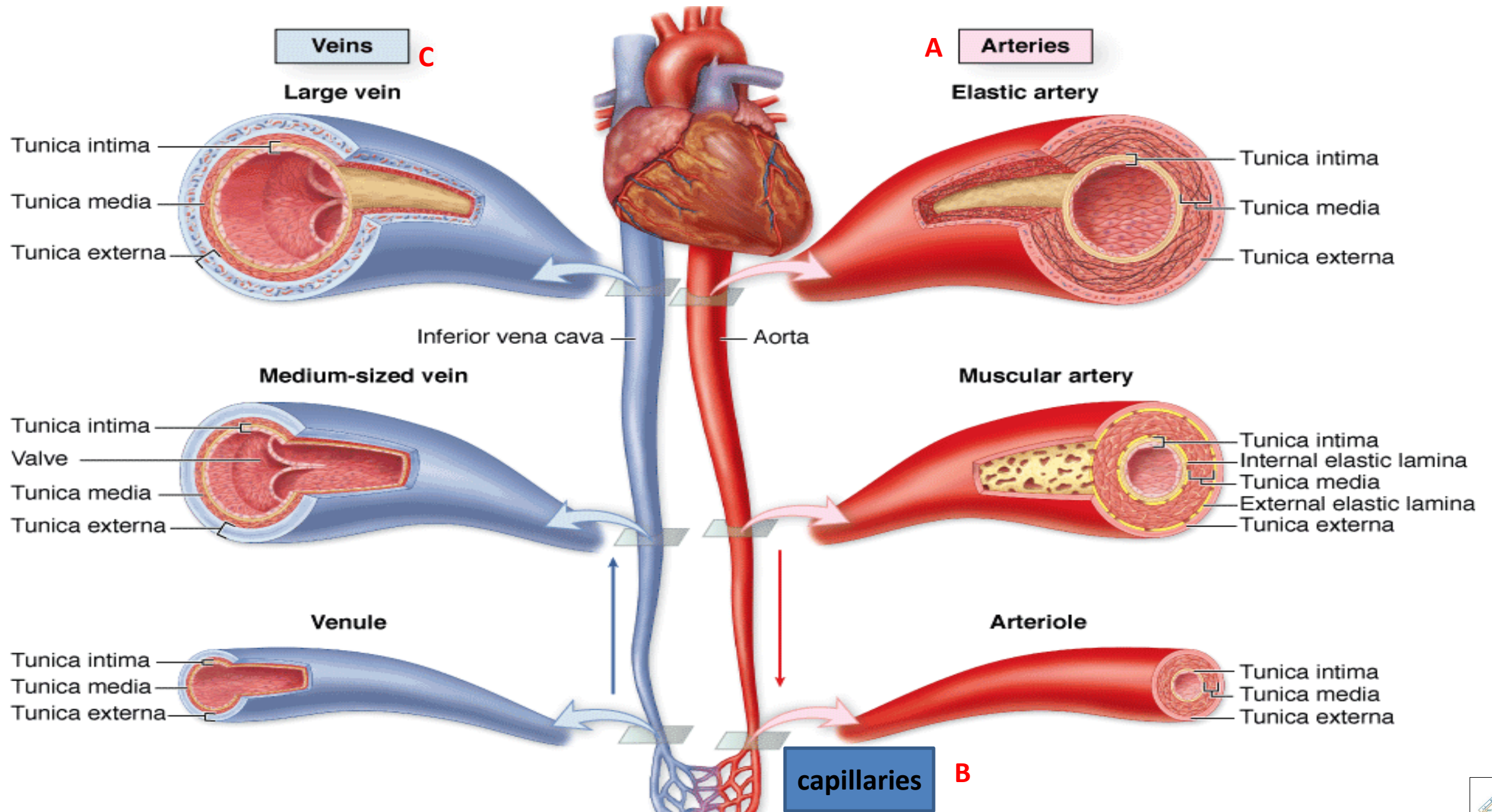
The circulatory system

pumps and directs blood cells and substances carried in blood to all tissues of the body. It includes both the blood and lymphatic vascular systems, and in an adult the total length of its vessels is estimated at between **100,000 and 150,000 kilometers**. The blood vascular system, or cardiovascular system consists of the heart and blood vessels. When the body is at rest, approximately 70% of the blood moves through the systemic circulation, about 18% through the pulmonary circulation, and 12% through the heart



Types of Blood Vessels





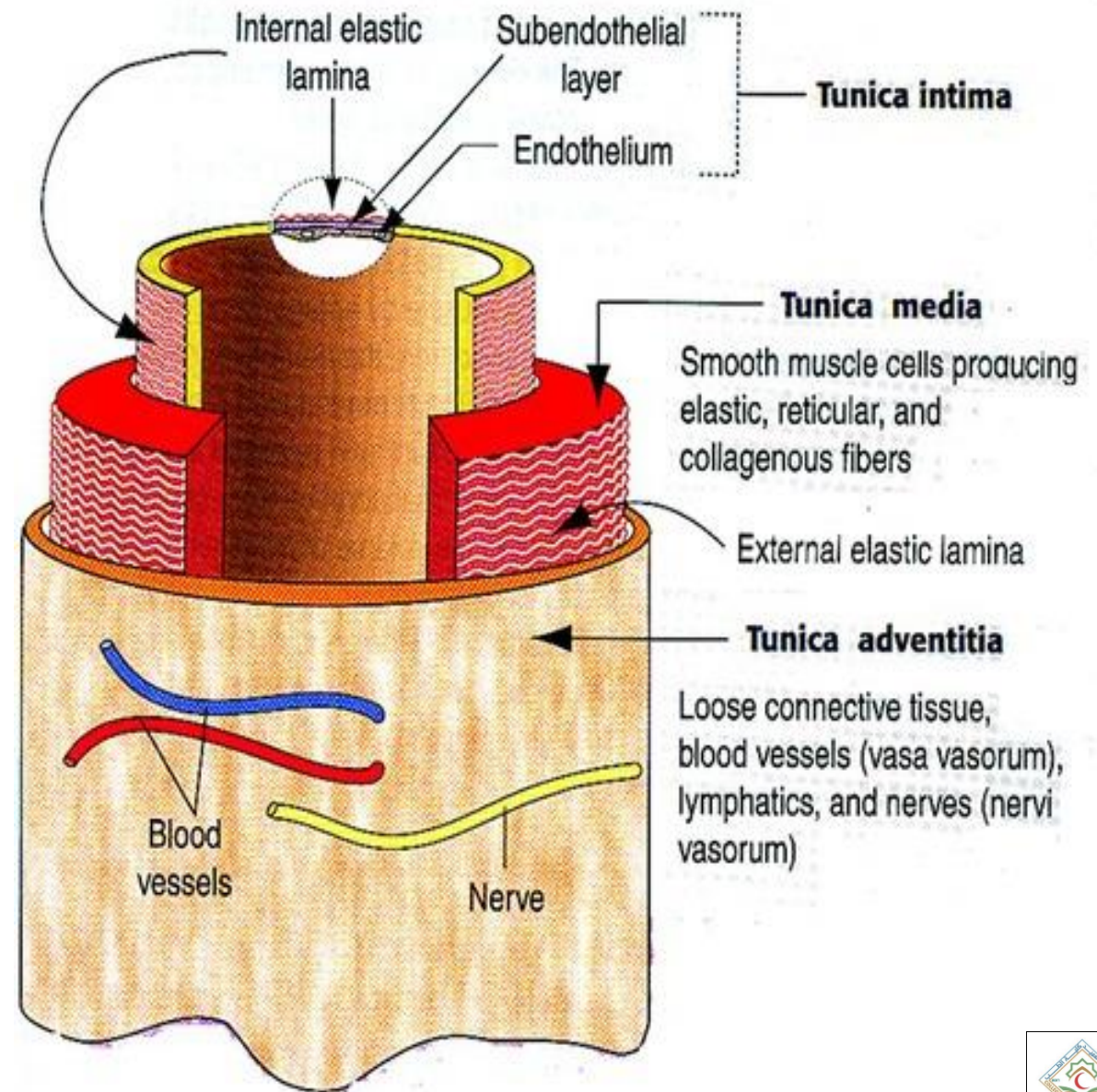
TISSUES OF THE VASCULAR WALL

The wall of blood vessels except capillaries is formed of three layers.

1.The tunica intima: the inner layer

2.The tunica media: the middle layer, its structure is related directly to the function of the vessel.

3.The tunica adventitia: or tunica externa is the outer layer.

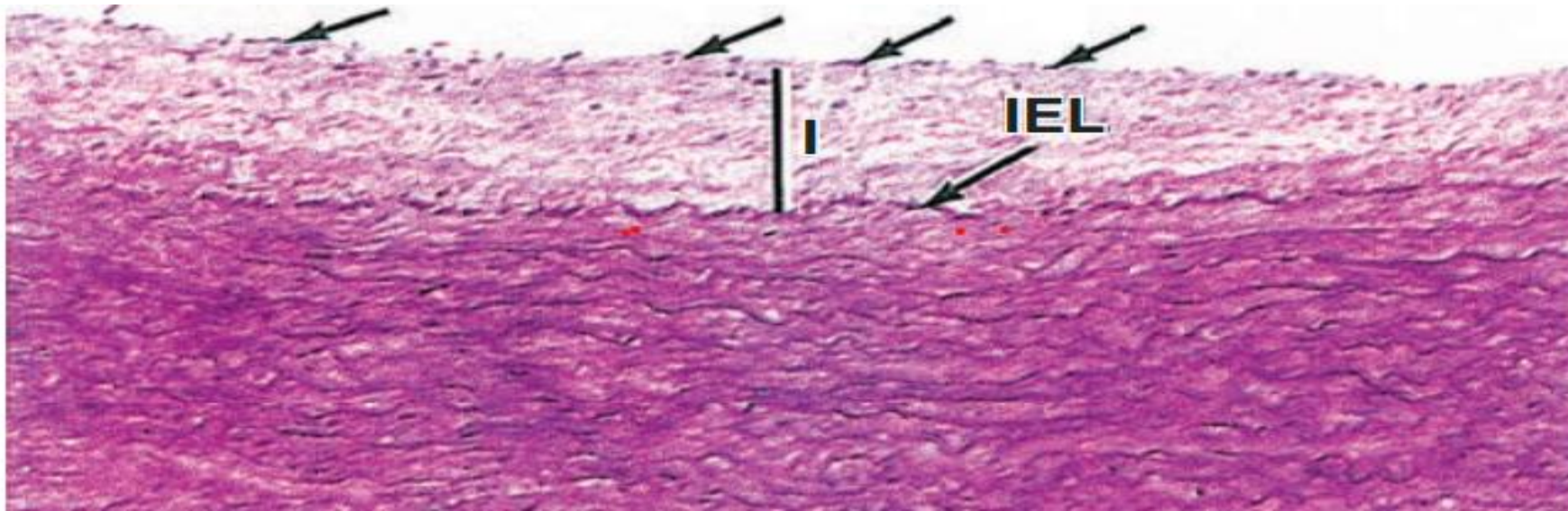


1.The tunica intima

A. Endothelium:

1. Vascular endothelial cells are squamous, polygonal, and elongated with the long axis in the direction of blood flow
2. Resting on a basal lamina
3. The cells are joined together by occluding junctions

B. Subendothelium: loose fibro-elastic C.T. In muscular arteries the intima includes a thin layer, **the internal elastic lamina**, composed of elastin, with holes allowing better diffusion of substances from blood deeper into the wall.



The endothelium is a special type of simple squamous epithelium

Functions

1. Acts as a **semipermeable barrier** between two internal compartments: the blood plasma and the interstitial tissue fluid
2. Production of **vasoactive factors** that affect the vascular tone, such as **nitric oxide**, **endothelins**, and vasoconstrictive agents
3. Conversion of circulating **angiotensin I to angiotensin II**
4. Contain unique very small, elongated vesicles called **Weibel-Palade bodies**, which contain **selectin and von Willebrand factor** involved in blood coagulation



MEDICAL APPLICATION

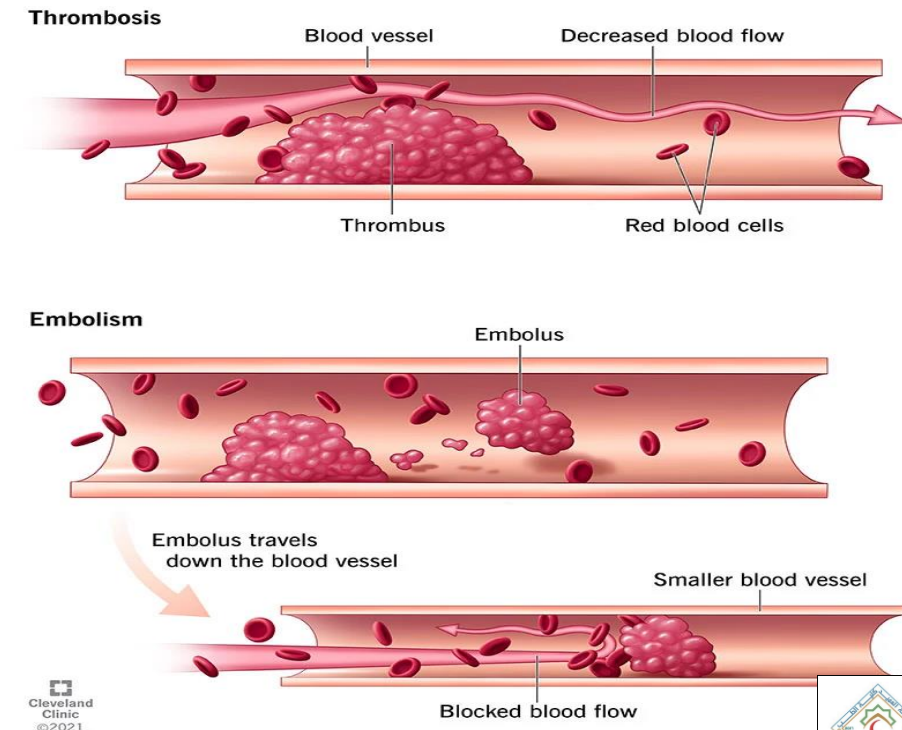
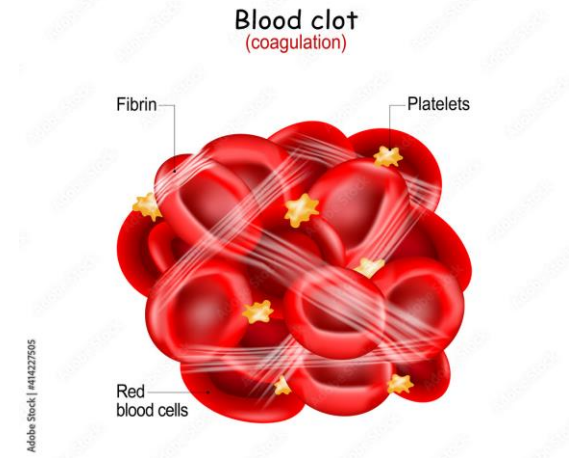
1. The endothelium also has **an antithrombogenic action, preventing blood coagulation**. When endothelial cells are damaged by atherosclerotic lesions, for example, the uncovered subendothelial connective tissue induces the aggregation of blood platelets.

2. This aggregation produces fibrin from circulating fibrinogen.

3. An intravascular clot, or **thrombus** is formed that may grow until there is complete obstruction of the local vascular flow.

4. From this thrombus, solid masses called **emboli** may detach and be carried by the blood to obstruct distant blood vessels. In both cases the vascular flow may stop, a potentially life-threatening condition.

Defective heart valves often replaced by an artificial valve. Because such valve replacements lack a complete endothelial covering, the patients require exogenous anticoagulant agents to prevent thrombus formation



2.The tunica media

A. **Concentric** layers of **helically** arranged smooth muscle cells.

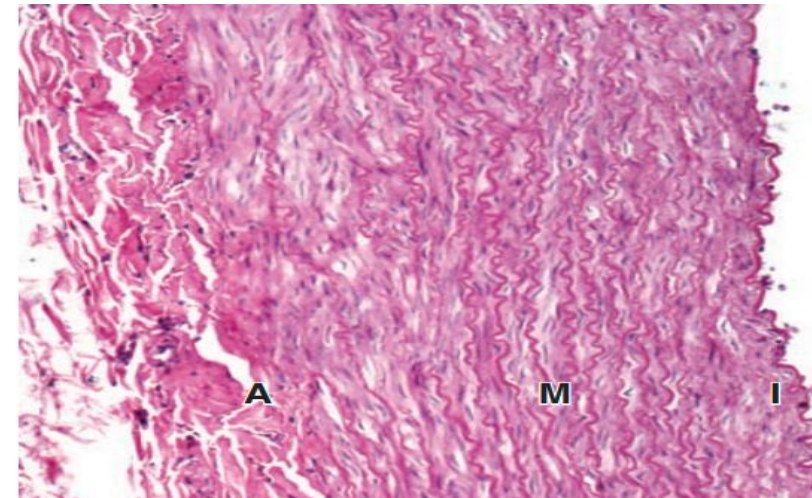
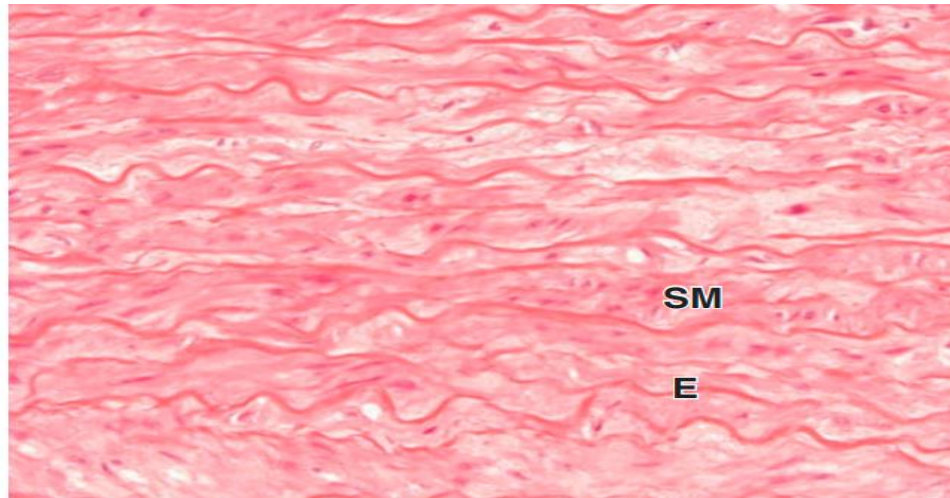
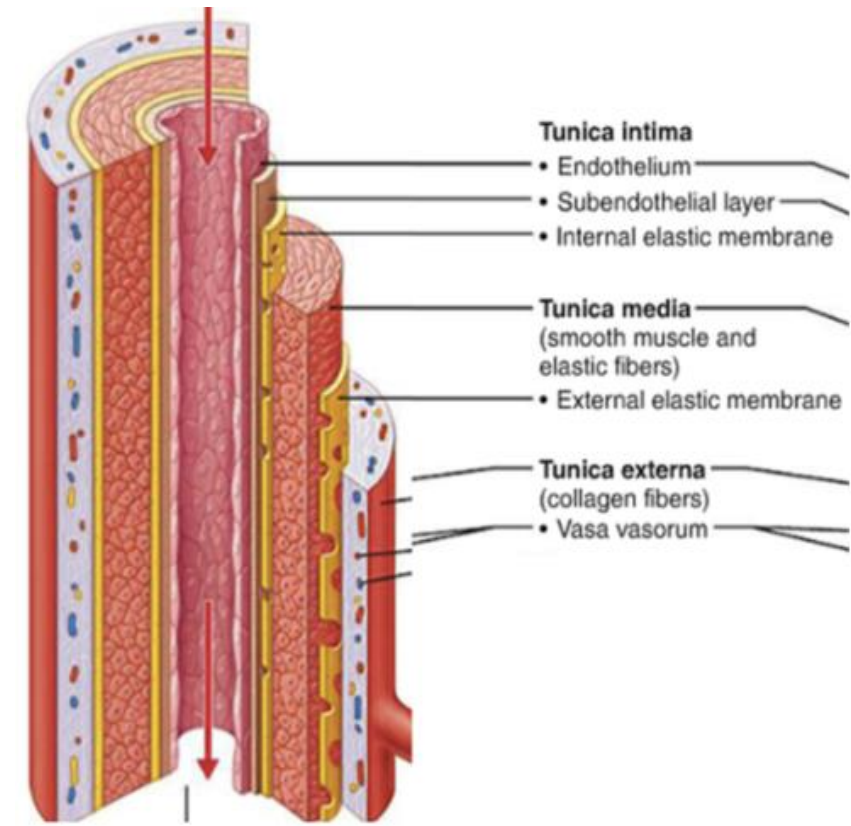
B. Elastic fibers.

C. Reticular fibers.

D. Extracellular matrix of proteoglycans.

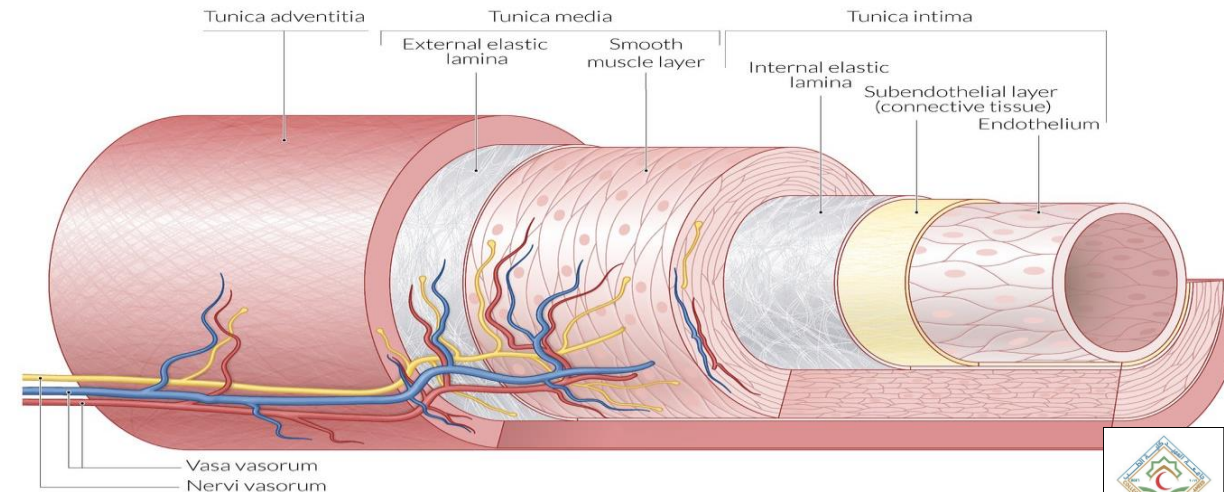
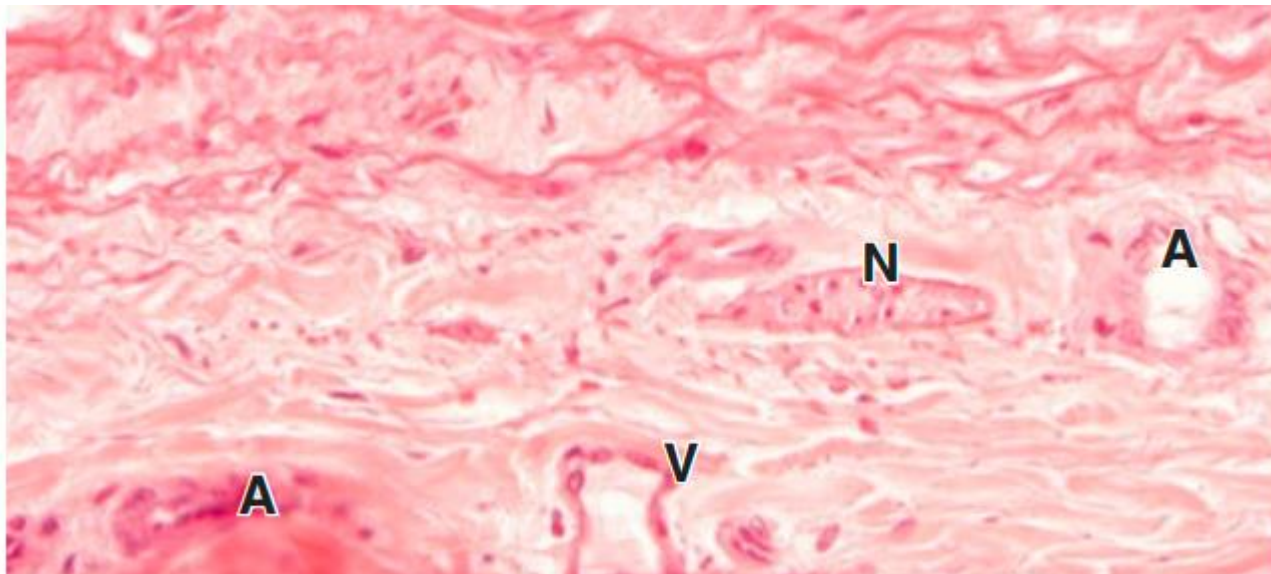
All of which are produced by the smooth muscle cells.

In arteries the media may also have an **external elastic lamina** separating it from the outermost tunica adventitia.



3.The tunica adventitia

- A. Fibroelastic connective tissue consisting principally of type I collagen and elastic fibers arranged longitudinally.
- B. Fibroblast.
- C. **Vasa vasorum** (“vessels of the vessel”): arterioles, capillaries, and venules in the adventitia and outer part of the media, provide nutrition to the wall of blood vessels
- D. **Nervi vasorum** : the vasomotor autonomic nerves



A. Arteries

- 1. Large Elastic Arteries
- 2. Muscular arteries
- 3. Arterioles

Basic Structure

Walls Contain
Three Layers

Tunica Intima

Endothelium
Basement Membrane
Delicate Connective T.

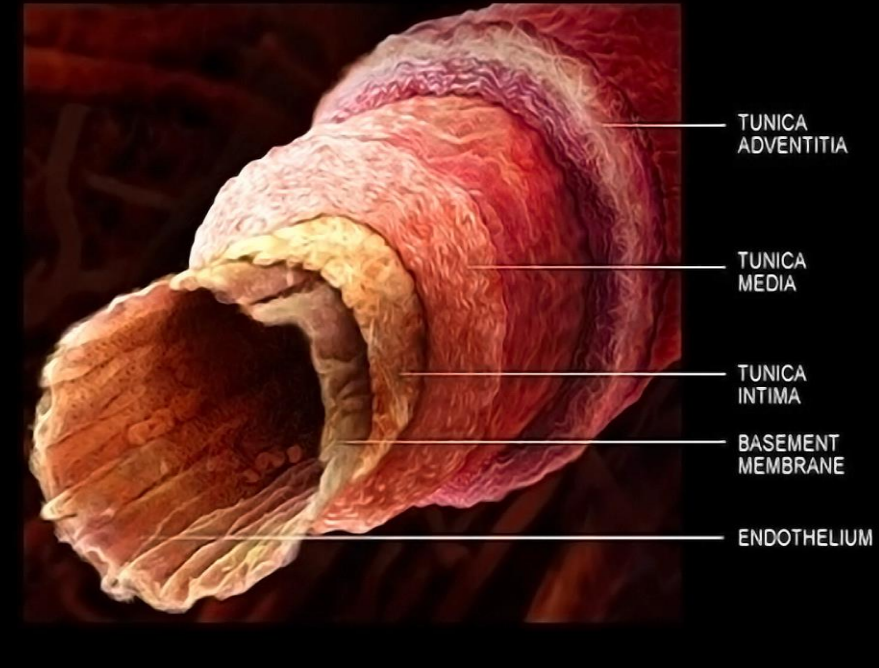
Tunica Media

Muscle / Elastic

Tunica Adventitia

Connective Tissue

An Artery's Layers



1. Large Elastic Arteries

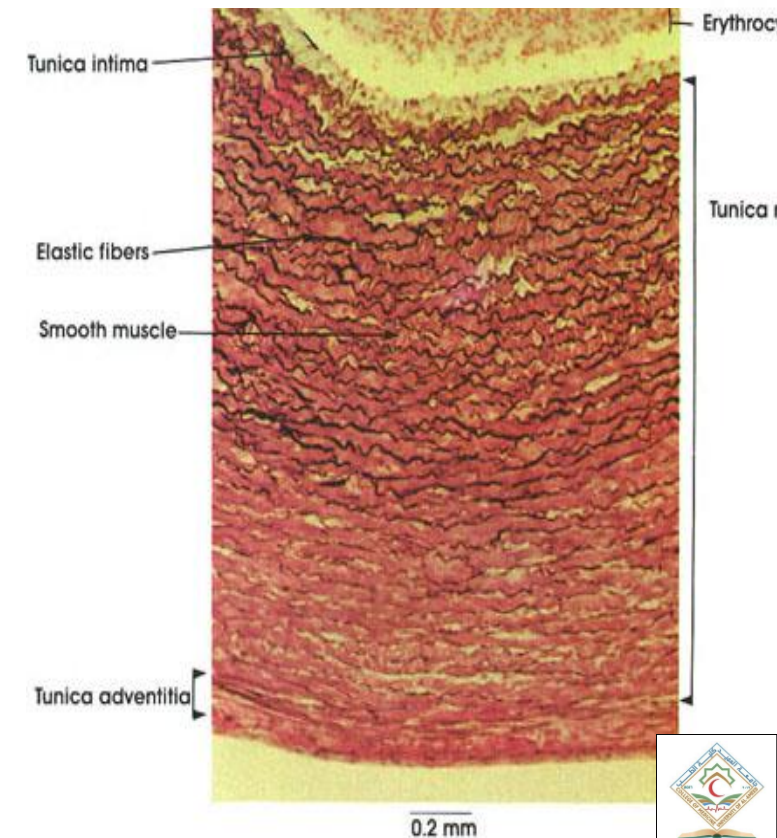
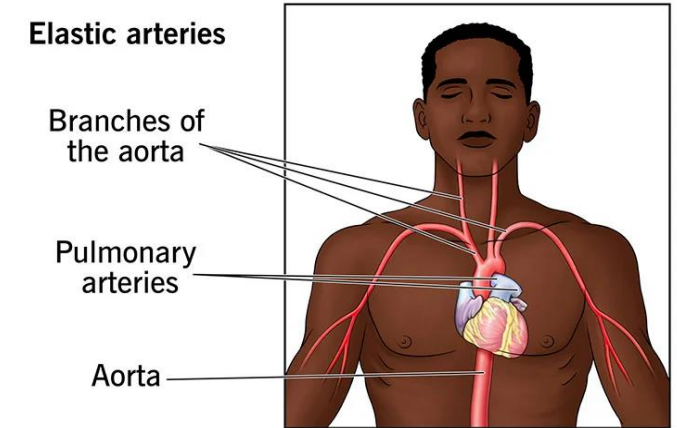
The elastic arteries include **the aorta** and its large branches.

A. The intima is thicker than the corresponding tunic of a muscular artery, forms 10% of the wall. The subendothelium contains also few smooth cells and macrophages.

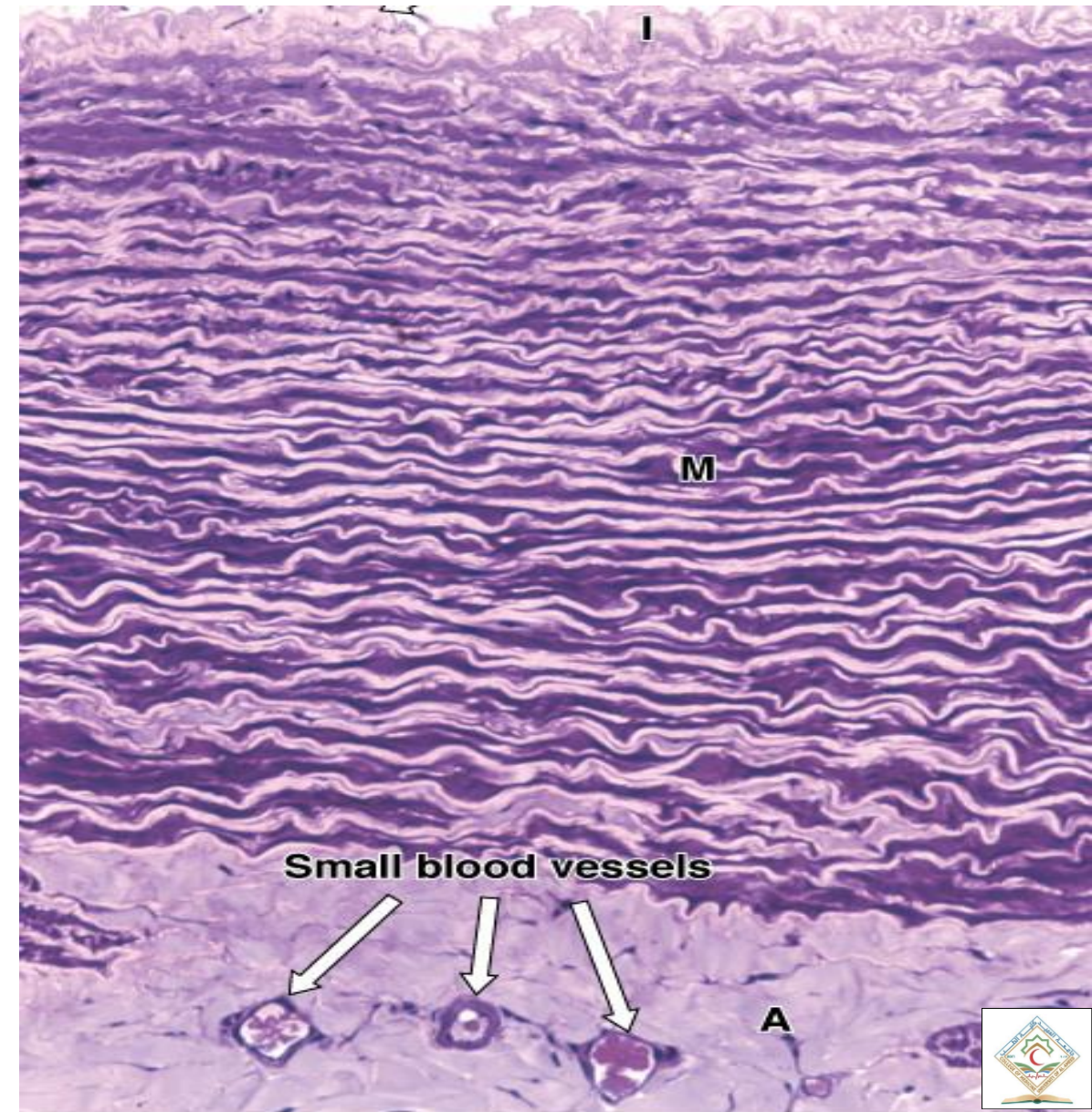
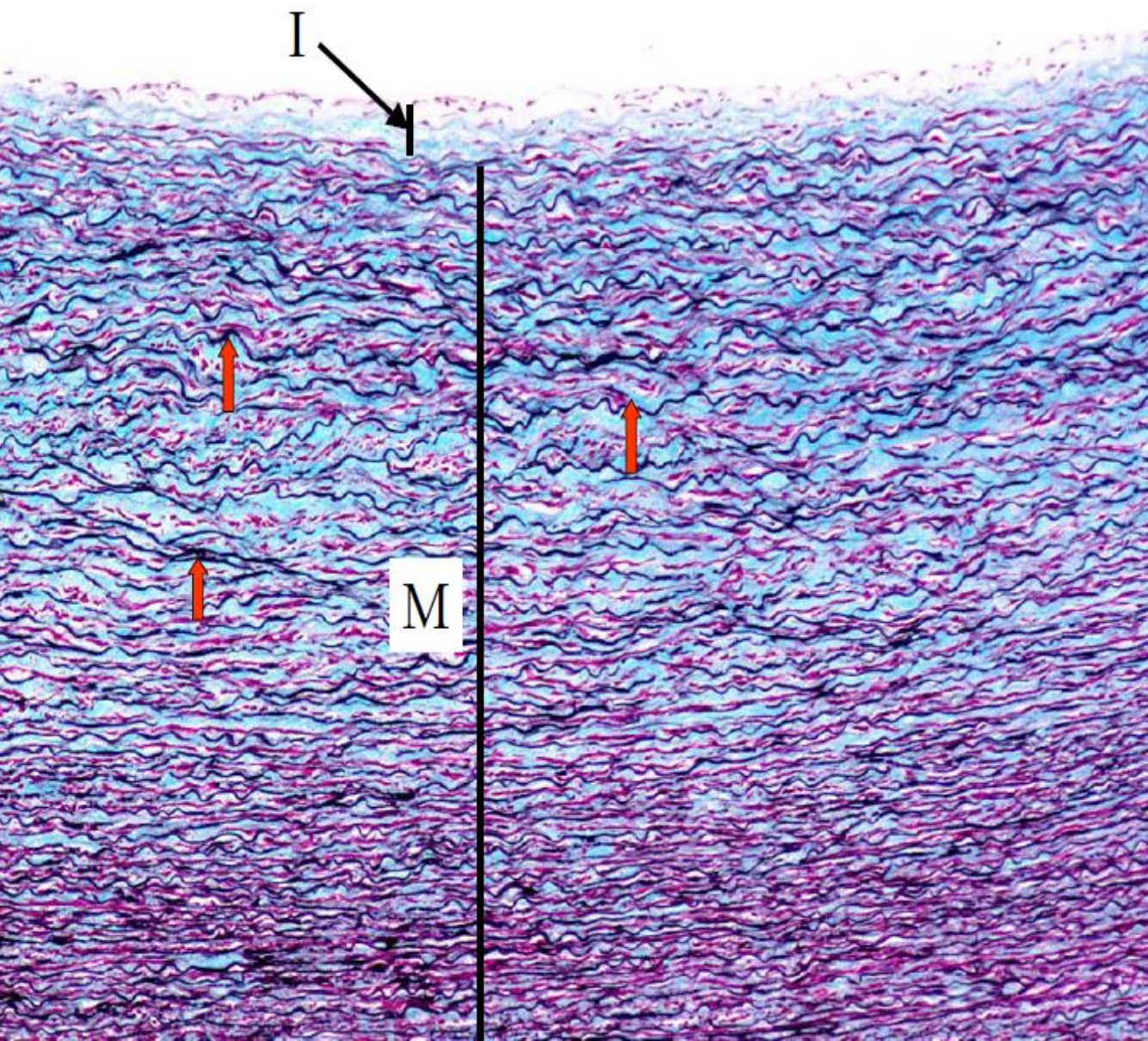
B. The media forms 70% of the wall and consists of elastic fibers and a **series of concentrically arranged, perforated elastic laminae** whose number increases with age (there are about 40 in the newborn, 70 in the adult). Between the elastic laminae are smooth muscle cells and reticular fibers.

C. The tunica adventitia is 20% of the wall. Vasa vasorum are numerous

Elastic Arteries



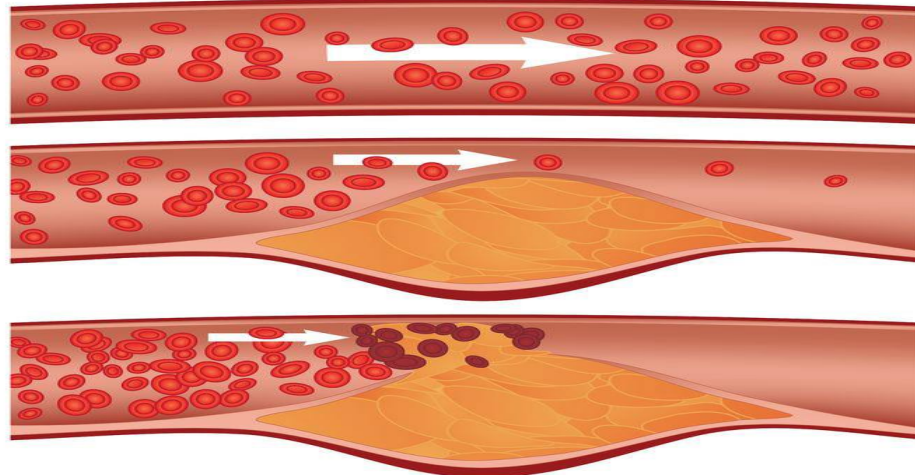
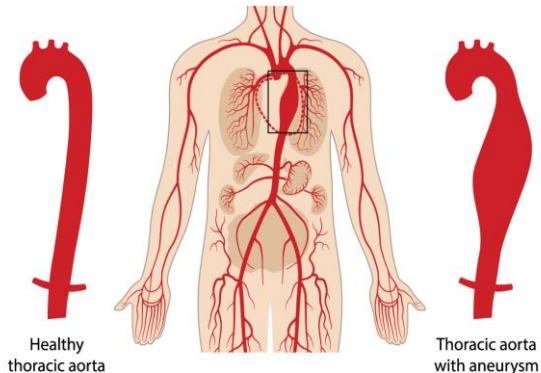
Large Elastic Arteries



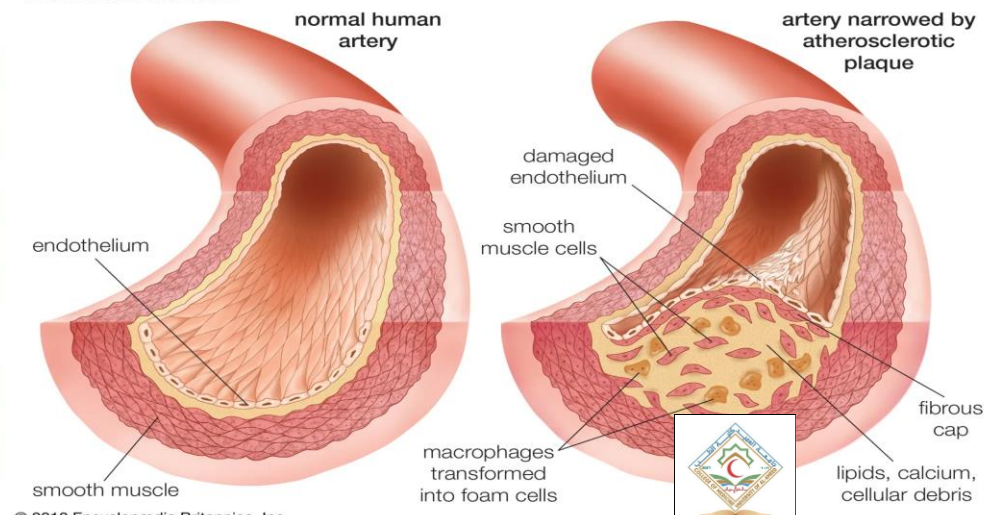
Atherosclerosis

- A. Is a disease of **elastic arteries and large muscular arteries** that may play a role in nearly half of all deaths in developed parts of the world.
- B. It is initiated by damaged of endothelial cells oxidizing **low-density lipoproteins (LDLs)** in the tunica intima, which induces adhesion and intima entry of **macrophages** to remove the modified LDL.
- C. Lipid-filled macrophages (called **foam cells**) accumulate and, along with the free **LDL**, produce a pathologic sign of early atherosclerosis called **fatty streaks**.
- D. During disease progression these develop into fibro-fatty **atheromas**, consisting of a gruel-like mix of **smooth muscle cells, collagen fibers, and lymphocytes with necrotic regions of lipid, debris, and foam cells**.
- E. In elastic arteries atheromas produce localized destruction within the wall, weakening it and causing arterial bulges or **aneurysms** which can rupture.
- F. In muscular arteries such as the coronary arteries, atheromas can occlude blood flow to downstream vessels, leading to **ischemic heart disease**

Thoracic Aortic Aneurysm



Atherosclerosis



2. Muscular arteries

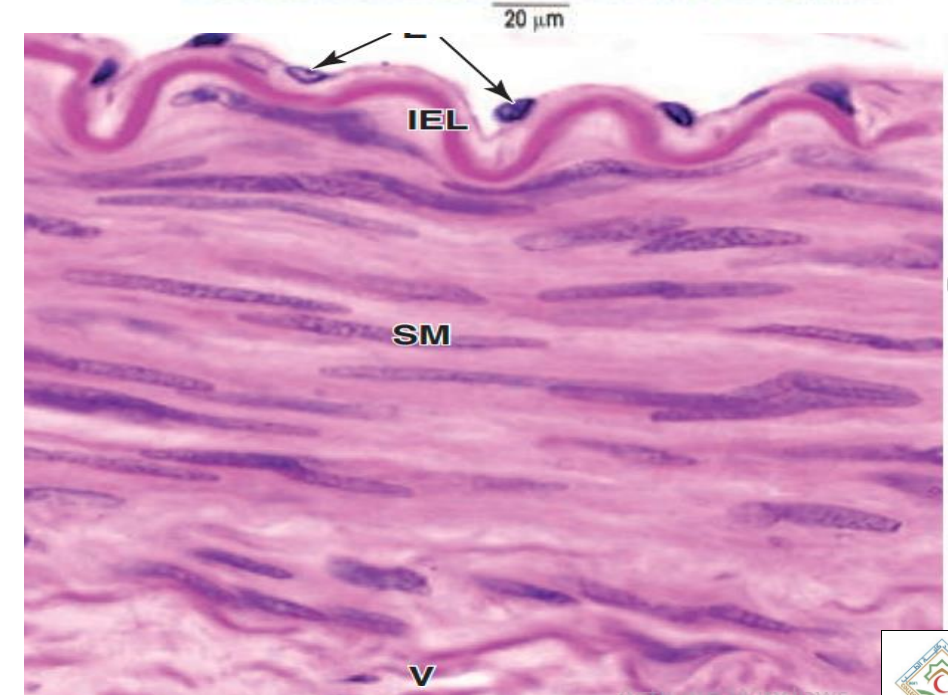
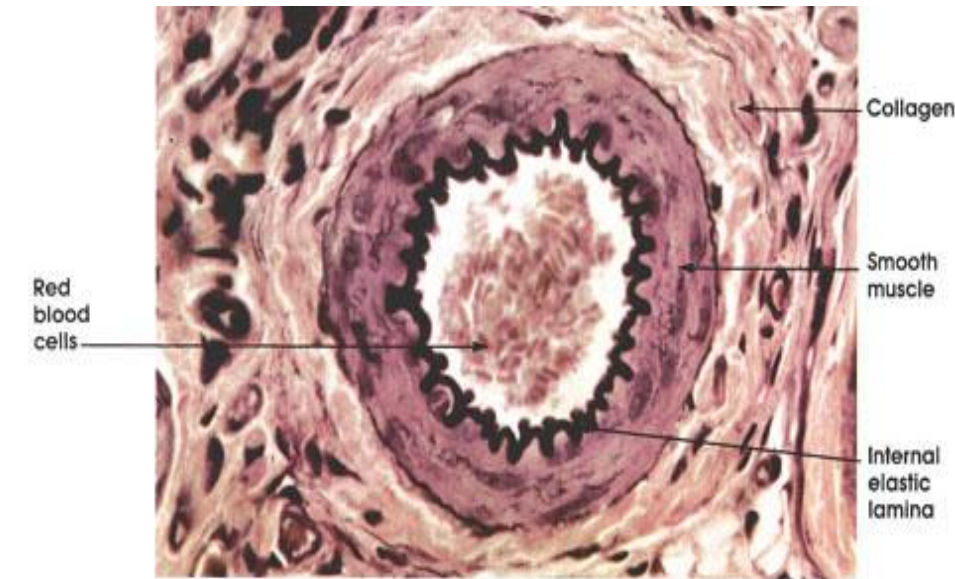
They include most of the **named arteries** as the femoral and ulnar arteries.

They have general structure of blood vessels and...

1.The intima has a very thin subendothelial layer and the internal elastic lamina, the most external component of the intima, is prominent

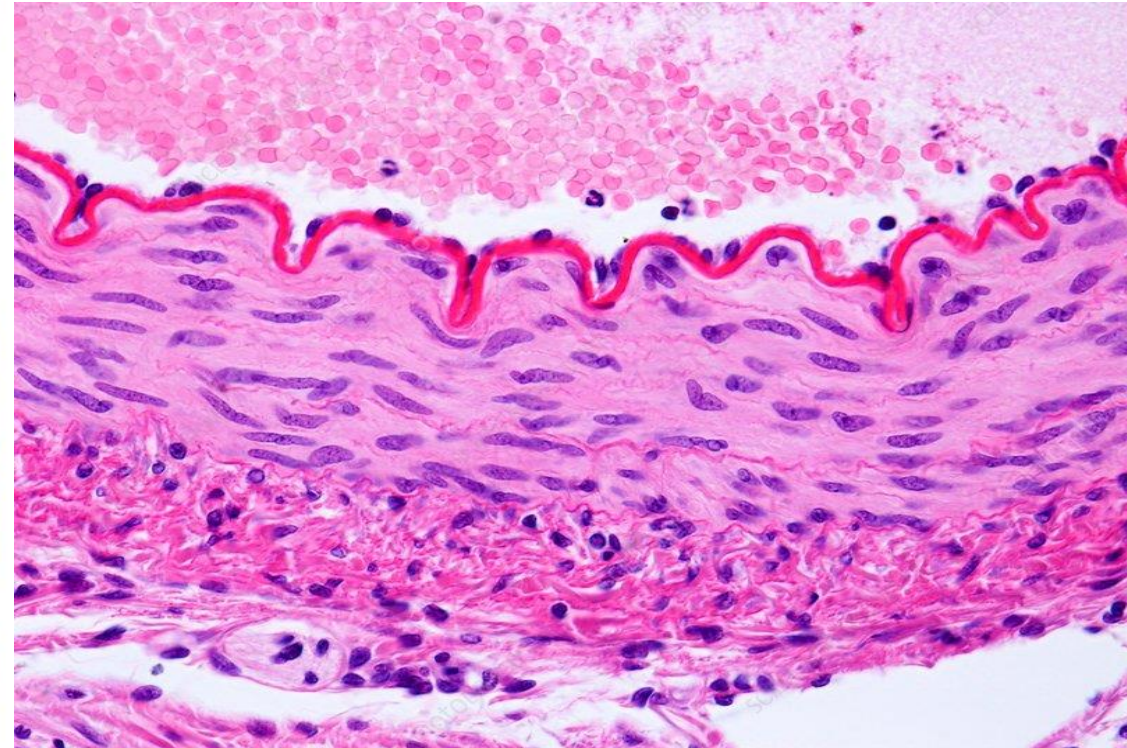
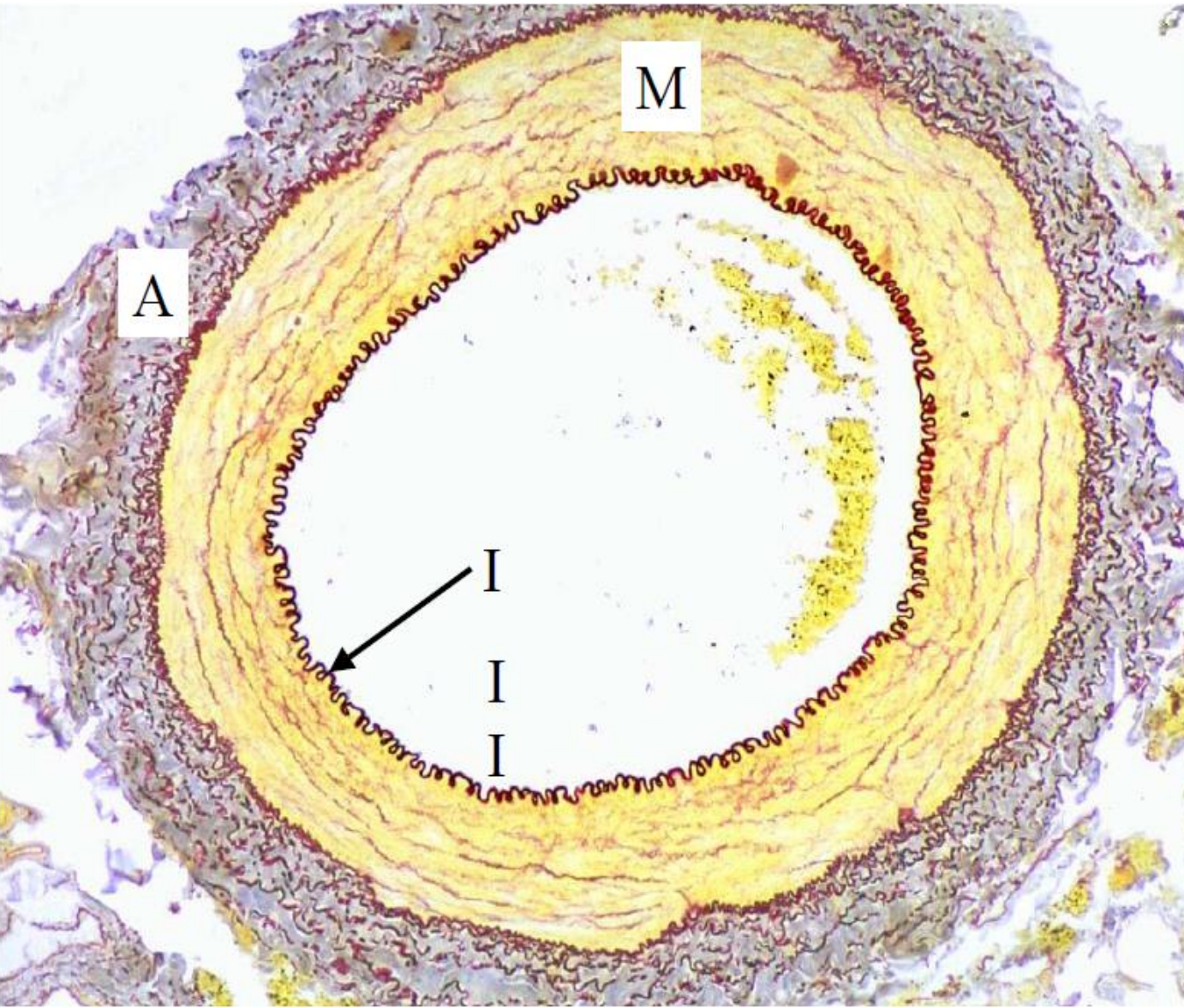
2.The tunica media may contain up to **40 layers** of more prominent **smooth muscle cells**. An external elastic lamina, the last component of the media, is present only in the larger muscular arteries. 50% of the wall

3.The adventitia consists of connective tissue. Lymphatic capillaries, vasavasorum and nervi vasorum. 50% of the wall



The muscular artery

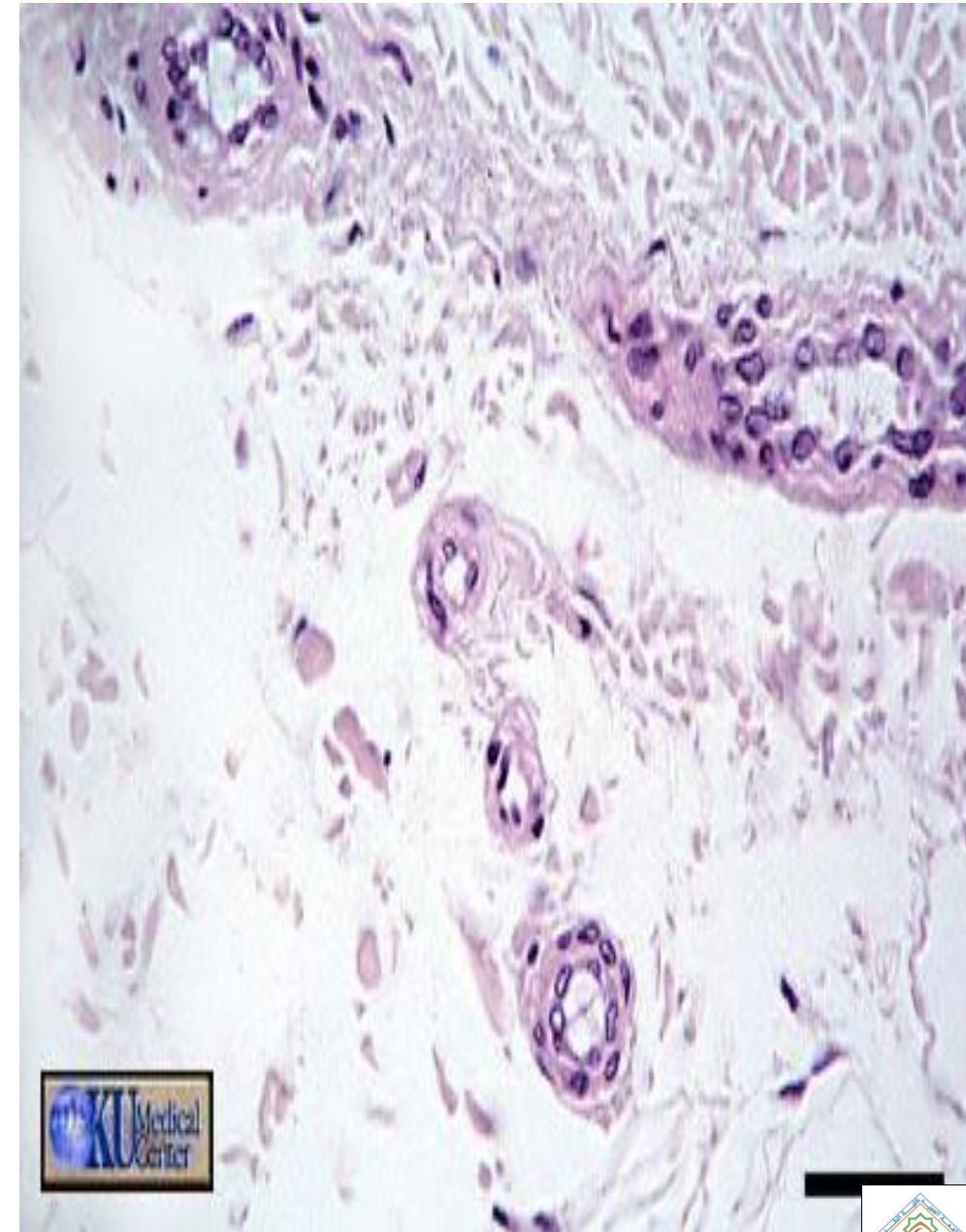
elastic elements



3. Arterioles

They control blood flow into capillaries.

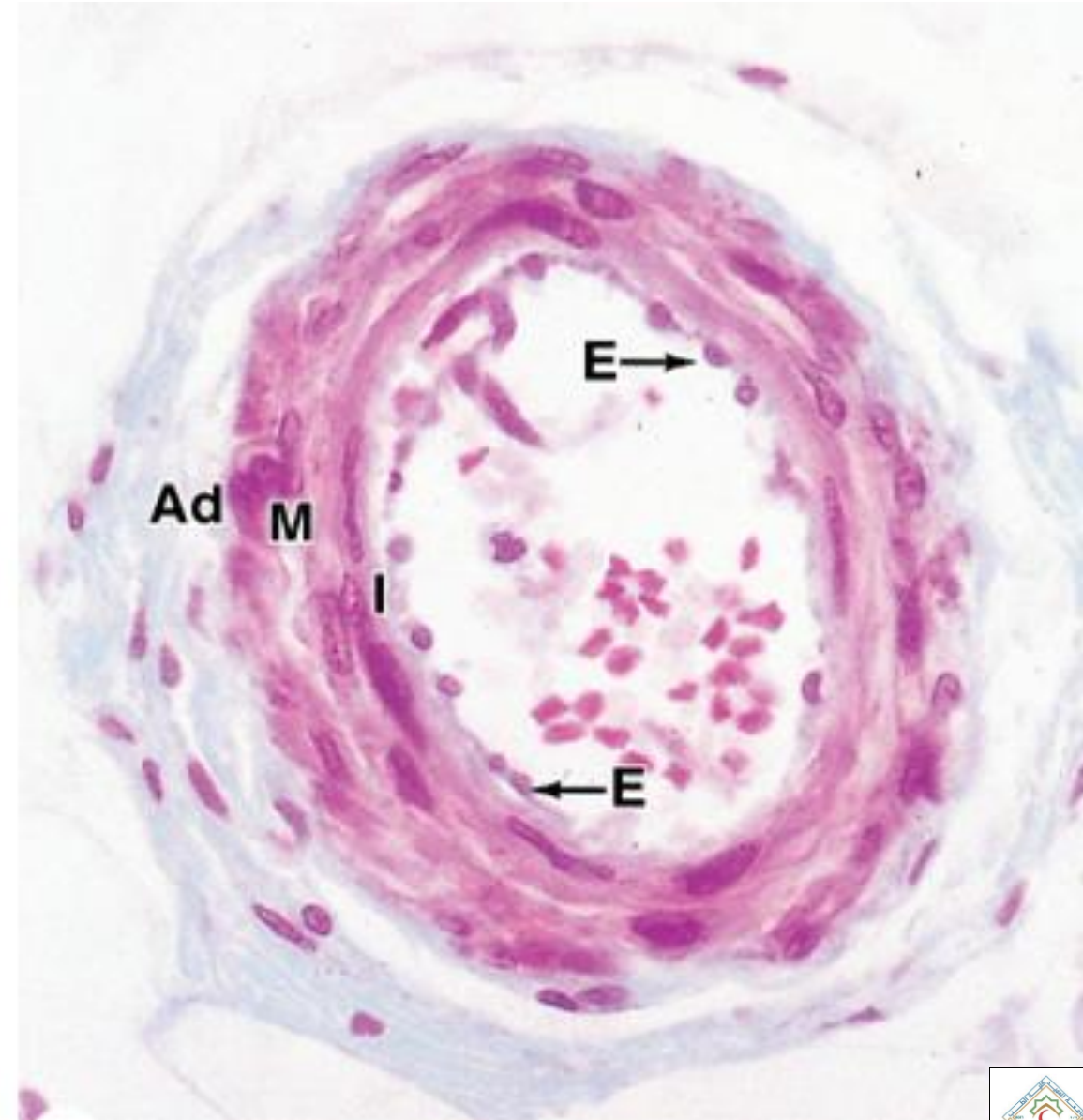
1. Are generally less than **0.1 mm** in diameter, with lumens approximately as wide as the wall is thick
2. The subendothelial layer is very thin, the elastic laminae are absent
3. The media is composed of **1-3 layers of circularly arranged smooth muscle cells.**
4. The tunica adventitia is very thin.



Arteriole

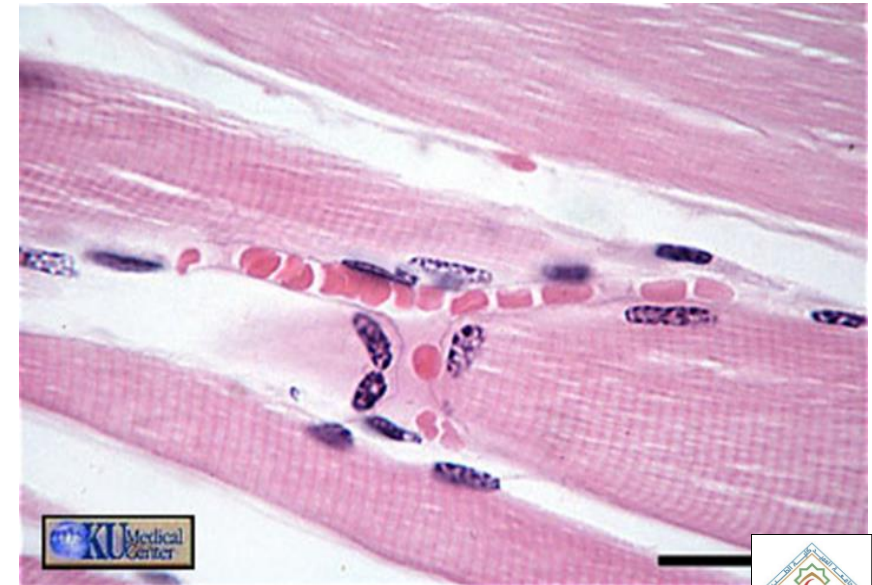
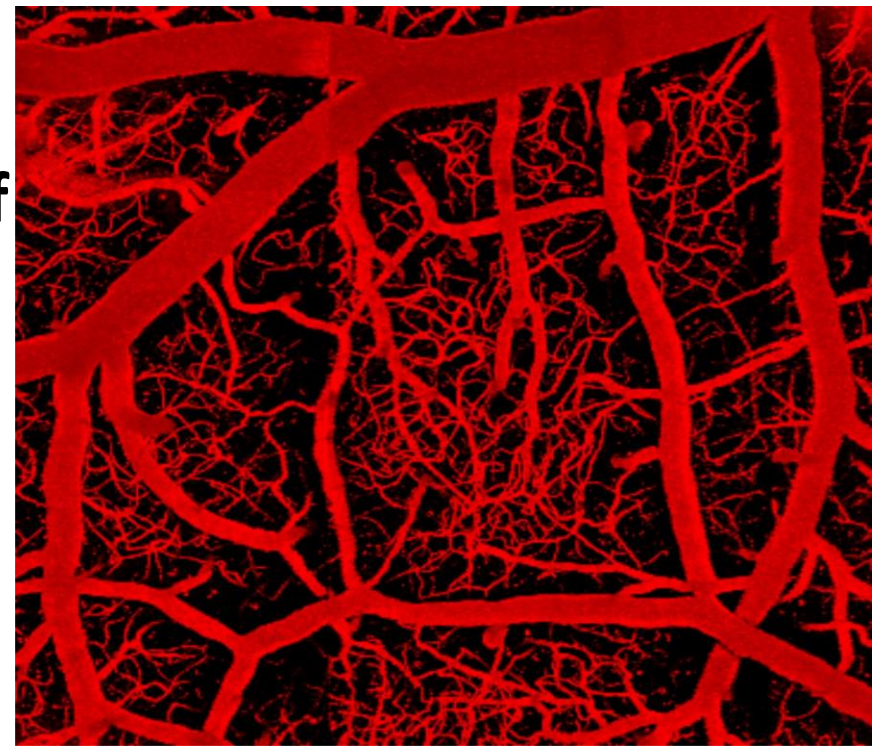
Blood pressure depends on cardiac output and the total peripheral resistance to blood flow, which is mostly due to the resistance of arterioles.

Essential hypertension, due to a wide variety of mechanisms that increase arteriolar constriction



B. Capillaries

1. The smallest blood vessels, permit different levels of metabolic exchange between blood and surrounding tissues.
2. They are composed of **a single layer of endothelial cells** rolled up in the form of a tube and rest on a basal lamina which is splitted to enclose the pericytes
3. The average diameter of capillaries varies from **4 to 10 μm** and their individual length is usually not more than 50 μm . Altogether capillaries comprise over 90% of all blood vessels in the body, with a total length of more than **100,000 km** and a total surface area of approximately **5000 m²**.

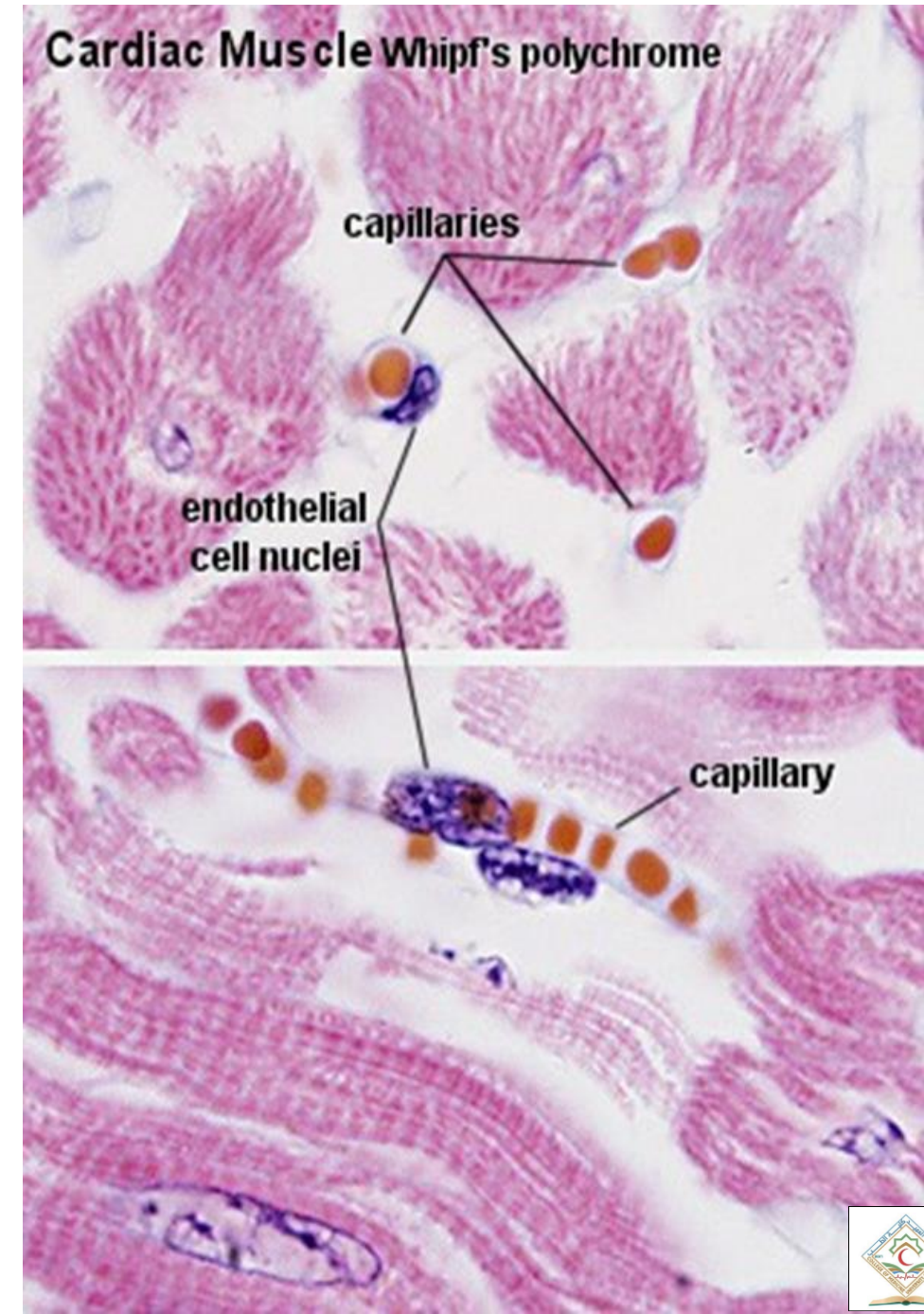


Capillaries

4. The total diameter of the capillaries is approximately 800 times larger than that of the aorta.

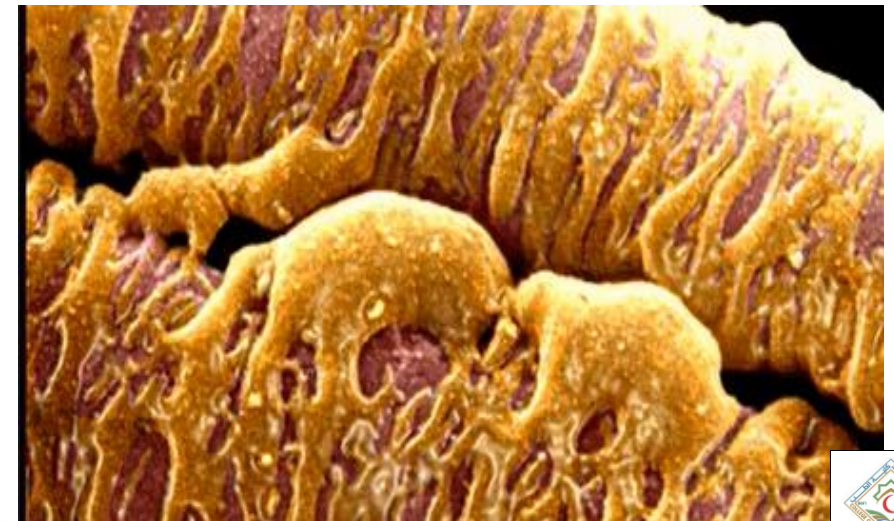
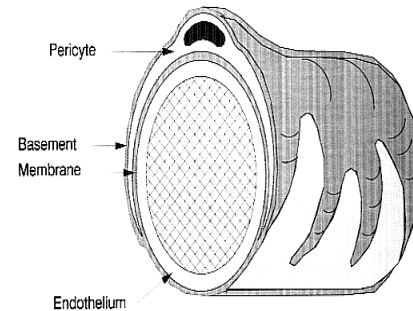
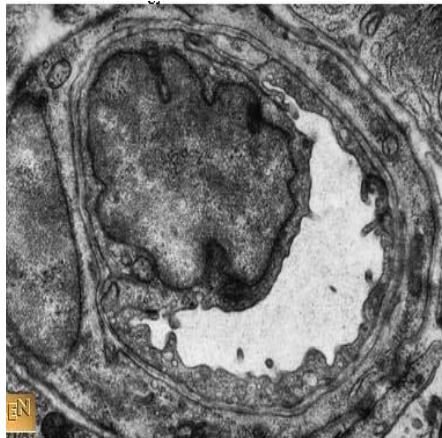
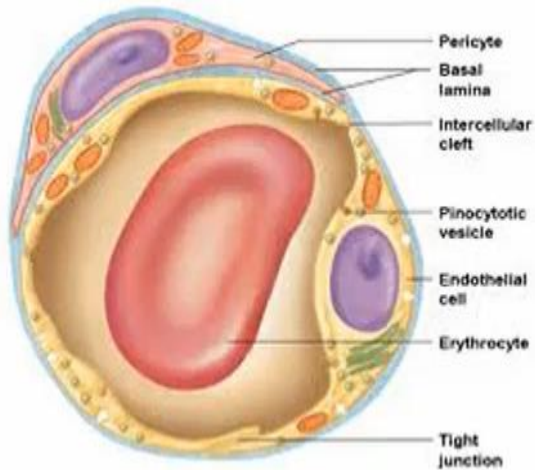
5. The velocity of blood in the aorta averages 320 mm/s, but in capillaries blood flows only about 0.3 mm/s. Because of their thin walls and slow blood flow, extensive surface area, and slow pulsatile blood flow capillaries are a favorable place for the exchange of water, solutes, and macromolecules between blood and tissues.

6. The excessive blood sugar that occurs with **diabetes** commonly leads to diabetic microangiopathy, **a diffuse thickening of capillary basal laminae** and concomitant decrease in metabolic exchange at these vessels, particularly in the kidneys, retina, skeletal muscle, and skin



Pericytes

1. Are perivascular cells of **mesenchymal origin** with **long cytoplasmic processes** partially surrounding the endothelial layer at various locations along **capillaries and postcapillary venules** and enclosed with the basal lamina of capillaries
2. The functions of these cells are
 - a. Primary have contractile function.
 - b. Support of capillaries and post capillary venules
 - c. After tissue injuries, pericytes proliferate and differentiate to form new blood vessels



CAPILLARY TYPES

Continuous
Capillary



Typical
Locations

fat
muscle
nervous
system

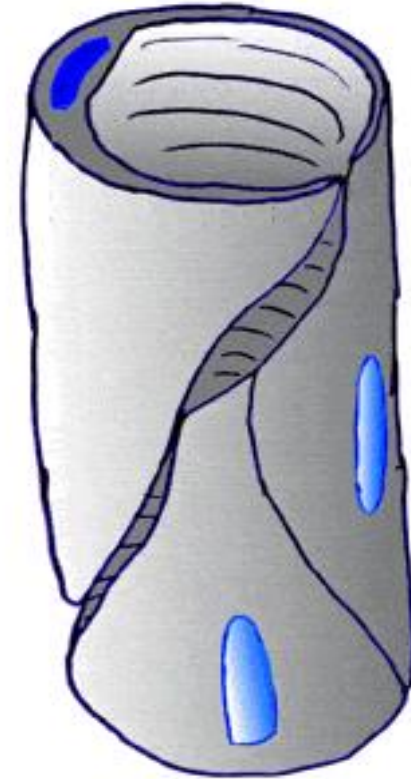
Fenestrated
Capillary



Typical
Locations

intestinal villi
endocrine glands
kidney glomeruli

Discontinuous
Capillary

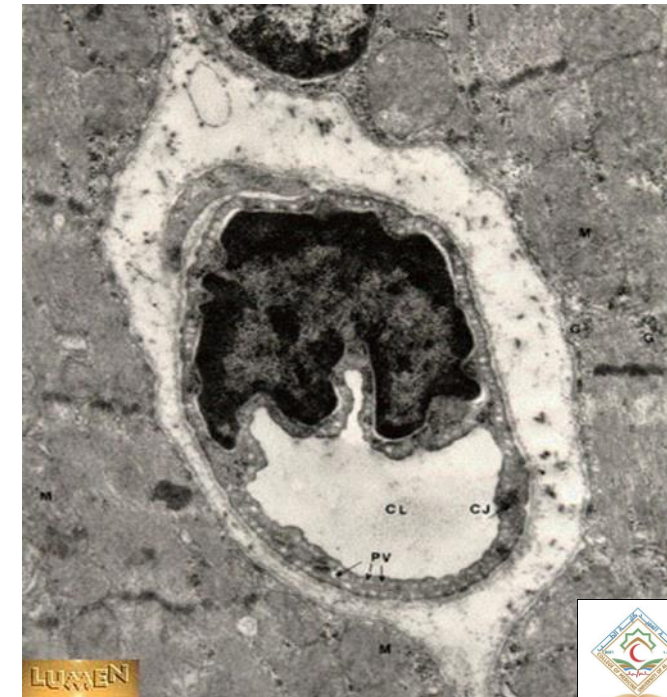
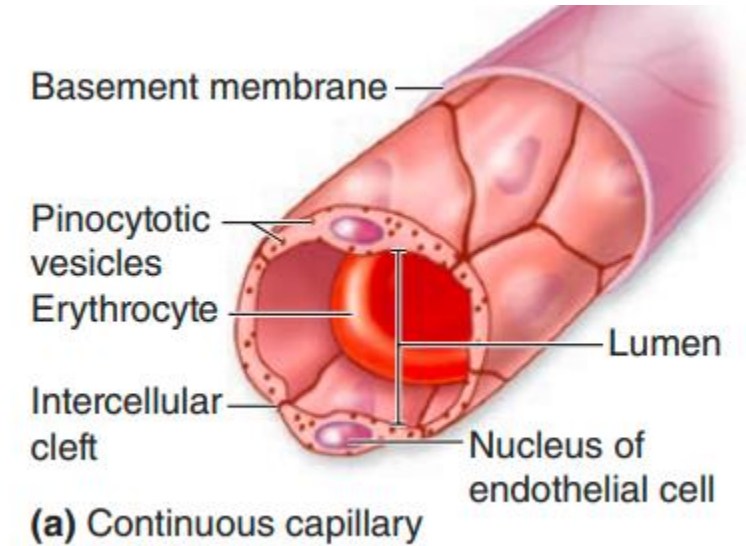


Typical
Locations

liver
bone marrow
spleen

1. Continuous capillaries

- a. The most common type, have **tight, occluding junctions** with overlapping marginal folds, sealing the intercellular clefts between all the endothelial cells to produce minimal fluid leakage.
- b. The basal lamina is continuous
- c. All molecules exchanged across the endothelium must cross the cells by **diffusion or transcytosis**.
- d. The cytoplasm contains many **pinocytotic vesicles**
- e. Found in all kinds of **muscle tissue, connective tissue, exocrine glands, lung and nervous tissue**
- f. Provide metabolic exchanges and share in the formation of different blood barriers (blood brain barrier, blood air barrier , blood thymus barrier)

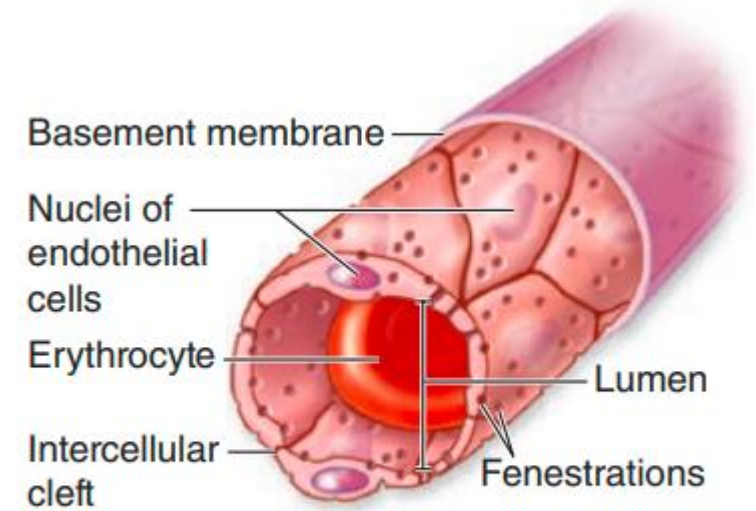


2. Fenestrated capillaries

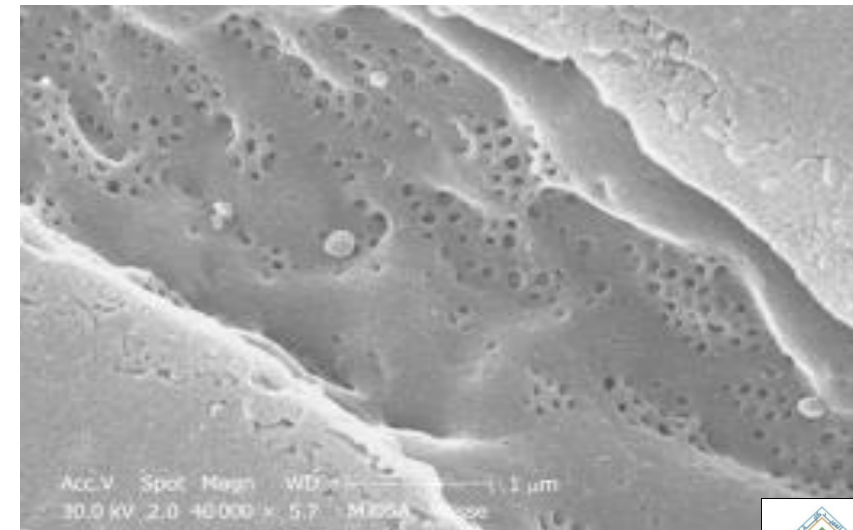
a. Also have **tight junctions** and the **basal lamina is continuous** but perforations (fenestrations) through the endothelial cells allow greater exchange across the endothelium.

b. Have a sieve like structure that allows more extensive molecular exchange across the endothelium. The endothelial cells are penetrated by numerous small circular openings or **fenestrations**. Some fenestrations are covered by very thin diaphragms of proteoglycans

c. Fenestrated capillaries are found in organs where molecular exchange with the blood is important, such as **endocrine organs, intestinal walls, kidneys and choroid plexus**

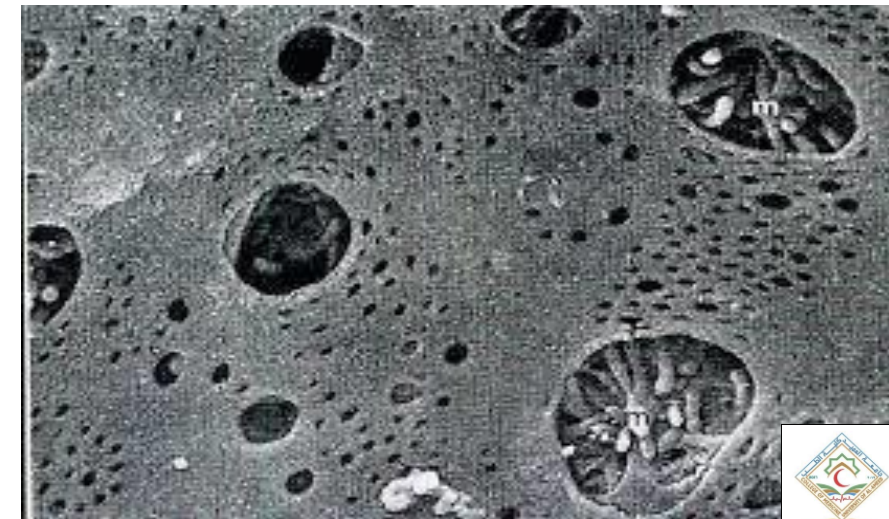
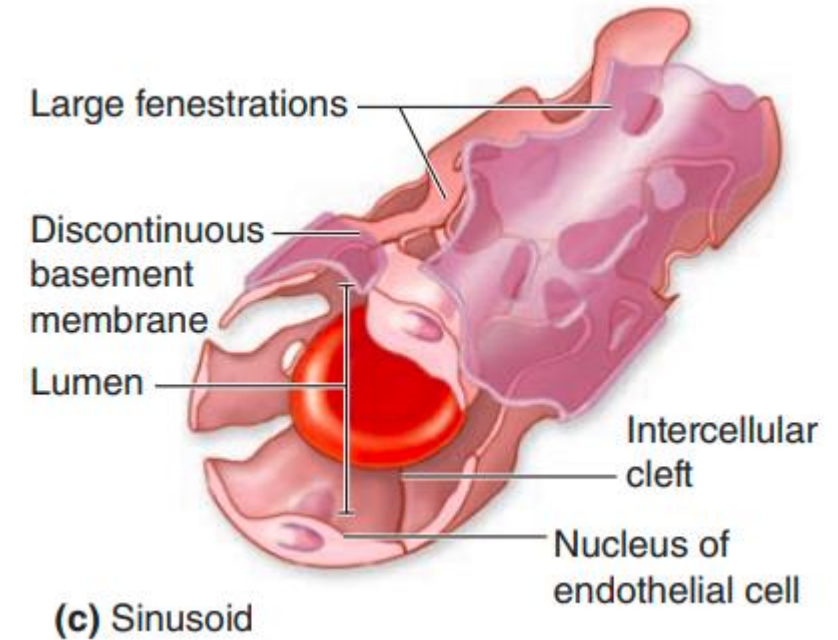


(b) Fenestrated capillary



3. Sinusoids, or discontinuous capillaries

- a. Capillaries with **large diameter** (30-40 μm) with irregular outline.
- b. The endothelial cells have large **fenestrations** with out diaphragms and are separated by **wide intercellular gaps**.
- c. The basal lamina is **discontinuous**.
- d. Macrophages are usually associated either within or around the sinusoids
- e. Allow maximal exchange of macromolecules and cells between tissue and blood
- f. Present in **bone marrow, liver, and spleen**



Thank
You



د. شهاب احمد فارس

شكرا

